

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

M. Tech. Electrical Semester – I, Examination, Dec 2013

EE704 Advanced Power System Protection

Date: 04/01/2014, Saturday

Time: 10.00 AM to 01.00 PM

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each section.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary

SECTION I

- Q - 1 (a) Explain advantages and special features that can be made available in numerical relay. [05]
- (b) Explain the term aliasing and hence give the solution of it [02]
- Q - 2 (a) Prove that the application of Least Square concept over a complete cycle results into the similar phasor components as obtained by DFT equation [07]
- (b) Draw and explain the block diagram for frequency insensitive phasor estimation using DFT [07]

OR

- Q - 2 (a) Explain various sinusoidal signal approximation based algorithm to compute the phasor with necessary derivations [07]
- (b) Draw and explain flowchart for an overcurrent relay [07]
- Q - 3 (a) Draw and explain flow chart for wavelet based transformer inrush current detection scheme [07]
- (b) Compare two sample, three sample and full cycle algorithm for phasor computation. [07]

OR

- Q - 3 (a) Write and explain steps for designing an IIR filter [07]
- (b) Explain Phasor measurement unit with necessary block diagram [07]

SECTION II

- Q - 4 (a) For the network shown in Fig. 1, find the primary-back up relay pairs for [07]
Relay No. 6 with the use of LINKNET structure.

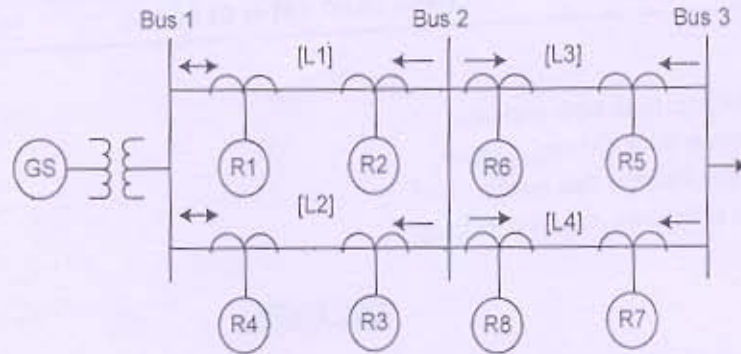


Fig. 1

- Q - 5 (a) Define power swing in power system. Derive the equation for the impedance seen [07]
by the distance relay in the event of power swing for two machine system. Show
the locus of power swing on the R-X plane with superimposed 3 stepped
characteristic of MHO relay.
- (b) Explain the need of out of step blocking and out of step tripping relay with an [07]
appropriate example. Explain the operation of out of step tripping and blocking
relay.

OR

- Q - 5 (a) [07]

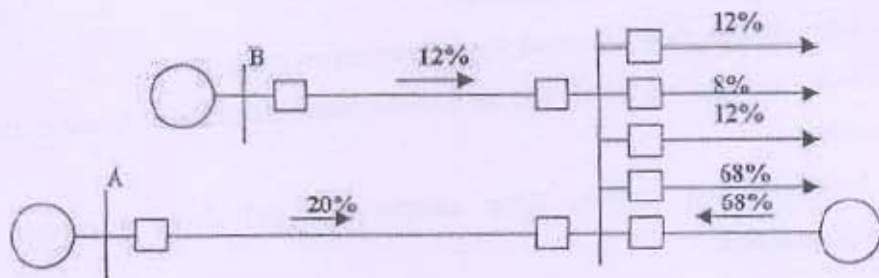


Fig. 2

Assume the first, second and third load-shedding frequencies are 59.5Hz, 58.9 Hz and 58.5 Hz, respectively. $H = 4$ sec. The relays have an effective 60-msec time delay and distribution breakers with an interrupting time of five cycles. Considering the system as shown in Fig. 2, find out the value of frequency at which generation and load are balanced after the worst overload case for loss of A, B or both.

- (b) Explain the working of KF Induction-Cylinder frequency relay. [04]
- (c) Compare the characteristic requirement for measurement and protective CT [03]
- Q - 6 (a) Explain different arrangement for protection of series capacitor banks with appropriate example [07]
- (b) Discuss the problem of reach measurement for distance relay for series compensated transmission line and explain under-reaching scheme. [07]

OR

- Q-6 (a) Discuss the technology infrastructure requirement for wide area measurement systems [07]
- (b) Explain the adaptive relaying with any one example for power system protection [07]
